

Service Module Test Plan

A complete, executable test plan for the Service module: equipment, contracts, job orders, planning, daily field work, materials, expenses, quotations, handover, invoicing, closeout, and reporting.

Every status value, closeout blocker, permission name, and menu path in this document was read from the code on 2026-07-26 rather than from older documentation.

Related documents:

- `docs/service-job-section-testing-document.md` — the longer narrative walkthrough of the same module
- `docs/testing-input-output-checklist.md` — field-by-field input/output values
- `docs/iss-tester-trainer-handbook.md` — environment setup and evidence rules
- `docs/end-to-end-testing-workflow.md` — the cross-module tally scenario

1. Before You Start

1.1 Environment

Local PostgreSQL on `localhost:5432`, database `iss`, credentials `pgadmin / vesper`. Do **not** use `docker compose up -d` for this — see `docs/iss-tester-trainer-handbook.md` section 6 for why.

```
dotnet run --project backend/src/ISS.Api/ISS.Api.csproj
cd frontend
npm run dev
```

`http://localhost:5257/health` must return `Healthy` before you begin. If it does not, stop and fix the environment; every failure after this point will be misleading.

1.2 Screen conventions

The Service module uses the modal pattern throughout. Create and edit forms open in a dialog over the

list; the list stays the primary view.

Element	Behaviour
+ New ... at the top right	Opens the create form in a dialog
Edit in a row	Opens a modal — the job header form, or the document editor for drafts
View in a row	Opens the full detail page
Delete	Browser confirmation first
Search box above a table	Filters rows already loaded on the page, not a server lookup

There is no `Cancel` button anywhere. Dismiss with `Close` , `Escape` , or a click outside. Dismissing must never save.

1.3 Roles

Run the first full pass as `Admin` . Most service job endpoints accept `Admin` , `Service` , and `Sales` ; `reopen` is restricted to `Admin` and `Service` . Re-run section 12 afterwards with a `Service` -only user.

1.4 Service menu

Menu item	Route
Command Center	<code>/service/command-center</code>
Dispatch Board	<code>/service/dispatch-board</code>
Technician Workbench	<code>/service/technician-workbench</code>
Equipment Units	<code>/service/equipment-units</code>
Service Contracts	<code>/service/contracts</code>
Job Orders	<code>/service/jobs</code>
Technicians	<code>/service/technicians</code>
Quotations	<code>/service/estimates</code>

Menu item	Route
Petty Cash	/service/expense-claims
Job Sheets / Work Orders	/service/work-orders
MRN	/service/material-requisitions
Inspection / QC	/service/quality-checks
Service Taken	/service/handovers

Job detail tabs: Overview , Plan , Daily Work , Materials , Expenses , Billing , Costs , Files & Notes . Sub-views: Daily Work → Daily Sheets / Staff / Labor / Progress ; Materials → Issued MRNs / Return Materials / Damage Material ; Expenses → IOU Advances / Petty Cash Expenses / Out-of-Pocket Claims .

2. Reference: Statuses

Testers should assert against these exact values. Anything else on screen is a defect.

Service job — Draft , Open , Assigned , InProgress , WaitingForParts , WaitingForCustomerApproval , WaitingForSupplier , WorkCompleted , PendingExpenseSettlement , PendingMaterialReturn , ReadyForInvoice , Invoiced , Closed , Reopened , Cancelled

Job kind — Service , Repair , Pdi , Warranty , Inspection

Entitlement source — None , ManufacturerWarranty , ServiceContract

Coverage scope — None , InspectionOnly , LaborOnly , PartsOnly , LaborAndParts

Billing treatment — Billable , PartiallyCovered , CoveredNoCharge

Entity	Statuses
Job operation (Plan)	Planned , InProgress , Completed , Skipped
Daily field sheet	Draft , Submitted , Approved , Rejected
Technician assignment	Pending , Approved , Rejected
Material requisition (MRN)	Draft , Posted , Voided

Entity	Statuses
Material disposition kind	Used , UnusedReturned , IncorrectReturned , Damaged , RejectedSupplierReturn
Material charge-to	Customer , Company , Supplier , Employee , Warranty
Work order	Open , InProgress , Done , Cancelled
Work order time entry	Draft , Submitted , Approved , Rejected , Invoiced
Expense claim	Draft , Submitted , Approved , Rejected , Settled
Expense funding source	OutOfPocket , PettyCash
Quotation (estimate)	Draft , Approved , Rejected
Quotation customer approval	NotSent , Pending , Approved , Rejected
Quotation line kind	Part , Labor , Expense
Service Taken (handover)	Draft , Completed , Cancelled

3. Reference: The 11 Closeout Blockers

A job refuses to close while any check is open. The API exposes them at `GET /api/service/jobs/{id}/closeout-checks` , and the `Billing` tab renders them.

#	Key	Label	Clears when
1	daily-field-sheets	Daily field sheets	Every daily sheet is approved or rejected
2	expense-claims	Expense claims	Every job expense claim is settled or rejected
3	petty-cash-iou	Petty cash IOUs	Every job IOU is settled, rejected, or cancelled
4	direct-purchase-bills	Direct purchase supplier bills	Every posted job-linked direct purchase has a posted supplier invoice

#	Key	Label	Clears when
5	material-requisitions	Draft material requisitions	No MRN is left in Draft (post or void)
6	job-assignments	Technician assignments	Every assignment is approved or rejected
7	labor-entries	Labor entries	No labor entry is left Draft or Submitted
8	work-orders	Job detail work orders	Every work order is Done or Cancelled
9	billable-labor	Uninvoiced billable labor	Approved billable labor is invoiced or marked non-billable
10	material-disposition	Material disposition	Every posted issue line is used/returned/damaged/supplier-returned
11	final-invoice	Final invoice decision	A final invoice exists, or the job is marked not billable with a reason

Section 10 is built around deliberately tripping and then clearing all eleven. That is the highest-value test in this module and the one most likely to catch a regression.

4. Test Data

Create this once. Later sections assume these codes.

Record	Value
Customer	CUS-SVC
Warehouse	MAIN
Equipment item (serial-tracked)	EQ-GEN01 , type Equipment , tracking Serial
Spare part item	SP-FILT , type SparePart , tracking None , cost 500
Sundries item	SP-GREASE , type SparePart , category SUNDRIES

Record	Value
Labour item	LAB-SVC , type Service
Equipment unit serial	SN-GEN-0001
Technician	TECH1 , cost rate 10 , billing rate 25
Opening stock at MAIN	SP-FILT qty 50 , SP-GREASE qty 20

Item Type accepts only Equipment , SparePart , Service . Tracking accepts only None , Serial , Batch . There is no "Stock", "Expense", or "Batch + Serial" option.

5. Equipment And Contracts

#	Step	Expected
5.1	Service -> Equipment Units , + New , register SN-GEN-0001 against EQ-GEN01 and CUS-SVC	Unit appears in list; serial unique
5.2	Re-register the same serial	Rejected with a clear duplicate message
5.3	Set warranty-until in the future	Unit shows warranty active
5.4	Service -> Service Contracts , create a contract for CUS-SVC covering the unit	Contract appears; coverage scope stored
5.5	Open the unit detail	Linked customer, contract, and job history visible

6. Job Order Lifecycle

#	Step	Expected
6.1	Service -> Job Orders , + New Job Order	Create dialog opens over the list

#	Step	Expected
6.2	Fill equipment SN-GEN-0001 , customer CUS-SVC , kind Repair , problem description, save	Dialog closes, job row appears without a browser reload
6.3	Note the job number	Format SJ.....
6.4	Edit on the job row	Job header edit modal opens directly from the list, not via the detail page
6.5	Dismiss with Escape after changing a field	Change discarded
6.6	Open the job, check Overview	Cockpit and process timeline render; status Open
6.7	Start	Status moves to InProgress
6.8	Complete	Confirmation dialog demands the literal word COMPLETE ; button stays disabled until typed exactly
6.9	Type complete in lowercase	Still disabled — the match is case-sensitive
6.10	Refresh entitlement	Entitlement source/coverage/billing treatment recalculated from warranty and contract
6.11	Reopen a closed job as Service role	Allowed
6.12	Reopen as Sales role	Denied — reopen is Admin and Service only

Entitlement matrix

Set up three jobs and confirm billing treatment derives correctly:

Unit state	Expected source	Expected treatment
In manufacturer warranty	ManufacturerWarranty	per coverage scope
Covered by active contract	ServiceContract	per coverage scope
Neither	None	Billable

Coverage LaborAndParts should give CoveredNoCharge ; a partial scope (LaborOnly , PartsOnly , InspectionOnly) should give PartiallyCovered .

7. Plan Tab

#	Step	Expected
7.1	Plan tab	Operations table is the primary content
7.2	+ Add Operation	Opens as a modal, not a passive card
7.3	Add three operations	All appear in sequence
7.4	Start an operation	Planned -> InProgress
7.5	Complete it	-> Completed
7.6	Skip another	-> Skipped
7.7	Delete an operation	Removed after confirmation

8. Daily Work

8.1 Daily sheets

#	Step	Expected
8.1.1	Daily Work -> Daily Sheets on an empty job	Clean empty state with + Create First Daily Sheet
8.1.2	Create a sheet	Modal; card appears with planned/done/pending counts
8.1.3	With records present	Header shows + Add Another Day
8.1.4	Submit	Draft -> Submitted
8.1.5	Approve	-> Approved

#	Step	Expected
8.1.6	Create a second sheet, Reject it	-> Rejected ; rejection reason captured
8.1.7	Daily sheet PDF	Downloads with the C-COM letterhead, header, planned/completed/pending, staff, progress, materials, IOUs, claims

8.2 Staff / Labor

#	Step	Expected
8.2.1	Open Staff / Labor with no sheet selected	Clean no-sheet message plus Go to Daily Sheets ; no disabled form
8.2.2	Assign TECH1 to a sheet	Assignment created as Pending
8.2.3	Approve	-> Approved
8.2.4	Create a second assignment, Reject	-> Rejected
8.2.5	Delete a pending assignment	Removed

Daily attendance is **not** the same as billable work-order labour. Confirm the two remain separate: a daily assignment must not create a billable time entry.

8.3 Progress

#	Step	Expected
8.3.1	Open Progress with no sheet selected	No-sheet message, no disabled form
8.3.2	Add a progress update	Appears in history, newest visible; history renders before the add form
8.3.3	Cockpit	"Last progress" reflects the new entry

9. Materials, Expenses, Labour, Quotation, Handover

9.1 MRN and stock

#	Step	Expected
9.1.1	Materials -> Issued MRNs , + New MRN	Modal opens
9.1.2	Add SP-FILT qty 4 from MAIN , post	MRN Draft -> Posted ; on-hand at MAIN drops 50 -> 46
9.1.3	Check the stock ledger	An issue movement is recorded against the job
9.1.4	Try to post more than on hand	Rejected with a clear message
9.1.5	For a serial item, post without selecting a serial	Rejected by serial validation
9.1.6	Void a posted MRN	Stock returns; status Voided
9.1.7	Leave one MRN in Draft	Needed for closeout blocker 5 in section 10

9.2 Disposition

Every posted issue line must end as Used , UnusedReturned , IncorrectReturned , Damaged , Or RejectedSupplierReturn , with a charge-to of Customer , Company , Supplier , Employee , Or Warranty .

#	Step	Expected
9.2.1	Mark 2 of 4 Used , charge Customer	Recorded; billable to the job
9.2.2	Return 1 as UnusedReturned	Stock at MAIN increases by 1
9.2.3	Mark 1 Damaged , charge Company	Recorded as a company cost, not customer-billable
9.2.4	On a separate line use RejectedSupplierReturn	Flows toward a supplier return
9.2.5	Leave one line undisposed	Needed for closeout blocker 10

9.3 Expenses

Three distinct instruments — confirm they stay distinct:

Instrument	Where	Funding source
IOU advance	Expenses -> IOU Advances	cash advanced before spend
Petty cash expense	Expenses -> Petty Cash Expenses	PettyCash
Out-of-pocket claim	Expenses -> Out-of-Pocket Claims	OutOfPocket

#	Step	Expected
9.3.1	Raise an IOU	IOU number confirmed; state says waiting for finance approval
9.3.2	Finance -> Petty Cash IOUs : approve, release from a fund, settle	Fund balance decreases on release; IOU reaches Settled
9.3.3	Raise an out-of-pocket claim, submit, approve, settle	Draft -> Submitted -> Approved -> Settled
9.3.4	Reject a second claim	-> Rejected
9.3.5	Convert billable claim lines to a quotation	Lines land on a service quotation
9.3.6	Leave one claim Submitted	Needed for closeout blocker 2

9.4 Work orders and billable labour

#	Step	Expected
9.4.1	Service -> Job Sheets / Work Orders , + New Job Sheet	Modal opens over the register
9.4.2	Add a time entry for TECH1 , 4 hours	Entry Draft ; cost uses 10 /h, billing 25 /h
9.4.3	Submit then approve	Draft -> Submitted -> Approved
9.4.4	Reject another entry	-> Rejected
9.4.5	Start then Done the work order	Open -> InProgress -> Done

#	Step	Expected
9.4.6	Leave one work order open	Needed for closeout blocker 8

9.5 Quotation

#	Step	Expected
9.5.1	Service -> Quotations , create for the job	Draft quotation
9.5.2	Add Part , Labor , and Expense lines	Totals recalculate
9.5.3	Send	Customer approval NotSent -> Pending
9.5.4	Approve	Quotation Approved
9.5.5	Revise an approved quotation	New revision; prior revision retained
9.5.6	Reject another quotation	-> Rejected
9.5.7	PDF	C-COM letterhead, correct lines and totals

9.6 Inspection / QC and Service Taken

#	Step	Expected
9.6.1	Inspection / QC , record a pass	Linked to the job
9.6.2	Record a fail with notes	Notes retained
9.6.3	Service Taken , create a handover	Draft
9.6.4	Complete	-> Completed
9.6.5	Convert to sales invoice with an approved quotation	Invoice created from quotation lines
9.6.6	On a second job, convert without an approved quotation, entering manual lines	Manual invoice supported: labour/work done, additional items, sundries, discount %, tax
9.6.7	Confirm the invoice in Sales -> Final Invoices	Present and linked back to the job
9.6.8	Try to create a handover for a closed job	Blocked with clear validation

#	Step	Expected
9.6.9	Cancel a draft handover	-> Cancelled

Grease and lubricants should use the `SUNDRIES` item category and must be invoiceable.

10. Closeout Gauntlet

The core regression test. Deliberately leave all eleven blockers open, then clear them one at a time.

Setup. On one job leave open: an unapproved daily sheet, a submitted expense claim, an unsettled IOU,

a job-linked direct purchase without a posted supplier invoice, a draft MRN, a pending assignment, a draft labor entry, an open work order, approved uninvoiced billable labour, an undisposed material line, and no final invoice decision.

#	Step	Expected
10.1	Open <code>Billing</code>	All 11 checks listed, each showing a pending count
10.2	Attempt <code>close</code>	Rejected. Message is Service job cannot close: <detail of the first open check>
10.3	Clear checks 1-10 one at a time, retrying <code>close</code> after each	Each retry fails on the <i>next</i> open check, never on one already cleared
10.4	Clear the final invoice decision by generating the invoice from the completed handover	Check 11 clears
10.5	<code>close</code>	Succeeds; status <code>Closed</code>

#	Step	Expected
10.6	Alternative for check 11: mark the job not billable with a reason	Check 11 clears without an invoice; reason is stored and visible
10.7	Reopen the closed job	Status Reopened ; job editable again
10.8	Re-close	Succeeds if nothing new was opened

Record which check each close attempt reported. A check that never surfaces, or one that blocks after

being cleared, is a defect worth logging in detail.

11. Costs, Dashboards, Reports

#	Step	Expected
11.1	Costs tab	Materials, labour, and expenses reconcile line-for-line with the MRNs, approved time entries, and settled claims
11.2	Compare against GET /api/service/jobs/{id}/costing	UI matches the API
11.3	Service -> Command Center	Active jobs, overdue jobs, jobs missing today's daily sheet, jobs missing today's progress, pending sheets, pending IOUs, pending claims, billing-ready handovers
11.4	Service -> Dispatch Board	Unassigned / assigned-active / waiting / completed lanes correct
11.5	Service -> Technician Workbench	Today's assignments and open daily sheets for the technician
11.6	Reporting -> Service KPIs	Counts agree with the jobs created in this run

#	Step	Expected
11.7	Reporting -> Costing	Job costs reflected in valuation
11.8	Job PDF	C-COM letterhead: logo, address, phone, email, website, and Page X of Y in the footer
11.9	Audit Logs	Status transitions recorded

12. Permissions

Re-run with a non-admin `Service` user, then with granular permissions removed. Each permission below

should hide the UI control **and** reject the API call — verify both. A hidden button with a still-open endpoint is a security defect, not a cosmetic one.

Permission	Denied user should not be able to
<code>Service.DailySheet.Submit / .Approve / .Reject</code>	Move a daily sheet out of Draft
<code>Service.JobAssignment.Create / .Approve / .Reject</code>	Assign or approve technicians
<code>Service.MaterialRequisition.Create / .Edit / .Post / .Void</code>	Create or post an MRN
<code>Service.ExpenseClaim.Create / .Submit / .Approve / .Reject / .Settle / .Convert</code>	Move a claim through its lifecycle
<code>Service.WorkOrderTimeEntry.Submit / .Approve / .Reject</code>	Approve billable labour
<code>Service.Estimate.Create / .Edit / .Approve / .Reject / .Send / .Revise</code>	Progress a quotation

Also confirm: a `sales` user can create and start jobs but **cannot** reopen one.

13. Negative And Edge Cases

#	Case	Expected
13.1	Issue material beyond on-hand	Rejected
13.2	Post a serial item without a serial	Rejected
13.3	Post the same serial to two jobs	Rejected
13.4	Close a job with any blocker open	Rejected, naming the blocker
13.5	Invoice a closed job	Blocked unless an authorised reopen path is used
13.6	Approve a daily sheet twice	Second attempt rejected or idempotent, never double-counted
13.7	Settle an IOU for more than advanced	Rejected or explicitly handled as extra reimbursement
13.8	Negative or zero quantities and hours	Rejected
13.9	Dismiss any modal mid-entry	Nothing saved
13.10	Two browser tabs editing the same job	Last write wins without corrupting status
13.11	Search a job number not on the current page	Will not be found — the search box filters loaded rows only. Expected behaviour, not a defect

14. Known Gaps

Not yet built. Do not raise these as defects — they are backlog, tracked in `docs/agent-handover-service-job-daily-operations.md` :

- job daily sheet report, technician daily work/time report, petty cash by job report
- IOU/employee advance report, employee reimbursement report
- material issued / returned / damaged / rejected by job reports
- pending daily sheet approval and pending job closeout reports
- daily sheet dashboard widget, mobile-friendly technician screen
- expense category master; attachments per expense line or daily sheet
- IOU outstanding balance reporting per employee/job

15. Sign-Off

Item	Value
Tester	
Date	
Environment	Local / Railway
Frontend commit	
API commit	
Job number used	
Sections passed	
Sections failed	
Closeout gauntlet (section 10)	Pass / Fail
Defects raised	

A run is only complete when section 10 passes end to end. Everything else can be partially covered; the closeout gauntlet cannot.